

REPORT OF Cybersecurity Internship: Development of Security-Based Applications

Submitted To:

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfillment of the requirement of degree of B.Tech CSE



BY:

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Batch: 2025-2029

CANDIDATE'S DECLARATION

I, **OM BHUSHAN** do hereby declare that the internship report titled "**Design and Development of Cybersecurity Tools and Secure Applications**" has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: OM BHUSHAN

Roll No: 25scs1003000995

Internship Completion Certificate::



4. PROJECT DETAILS

4.1 Introduction:

The cybersecurity internship provided practical exposure to different areas of cybersecurity and secure software development. During the internship, four projects were completed: **Password Strength Analyzer, Vulnerability Scanner, Phishing Email Detection, and Secure Login System.**

These projects helped in understanding important concepts such as password security, network scanning, phishing detection, machine learning, secure authentication, password hashing, SQL injection prevention, and session management.

4.2 Organization Profile:

Organization Name: Thiranex

Thiranex provides internship opportunities that allow students to gain practical experience through project-based learning. The internship focuses on developing technical knowledge and practical skills by working on real-world projects.

During the cybersecurity internship, multiple tasks were assigned to develop a better understanding of cybersecurity concepts and their practical implementation.

4.3 Problem Statement

Cybersecurity threats such as weak passwords, open network ports, phishing emails, and insecure login systems can lead to unauthorized access and data loss.

The purpose of these projects was to develop simple solutions that address these common security problems:

1. Analyze the strength of passwords.
2. Identify open ports and basic security risks.
3. Detect phishing emails using machine learning.
4. Develop a secure user authentication system.

4.4 Project Objectives

The main objectives of the internship projects were:

- To understand password security and password strength analysis.
- To learn basic network and port scanning.
- To identify potential vulnerabilities in network services.
- To understand phishing attacks and email classification.
- To apply machine learning concepts in cybersecurity.
- To develop a secure login system.
- To implement password hashing using bcrypt.
- To understand SQL injection prevention.
- To implement session management and logout functionality.

4.5 Scope of the Project

The scope of this internship project is focused on the development and implementation of basic cybersecurity tools and secure applications. The project covers four different areas of cybersecurity: **password security, network vulnerability assessment, phishing detection, and secure user authentication.**

The projects are designed to provide practical understanding of common cybersecurity problems and demonstrate how programming can be used to develop simple solutions.

1. Password Strength Analyzer

This project analyzes the strength of a password based on different security factors such as password length, uppercase letters, lowercase letters, numbers, and special characters. Based on these factors, the password is classified as **Weak, Medium, or Strong**. The project helps users understand the importance of creating strong passwords to reduce the risk of unauthorized access.

2. Vulnerability Scanner

The vulnerability scanner focuses on identifying open ports on a target system. It checks commonly used ports and provides basic information about potentially insecure services. The project helps in understanding basic concepts of **network security, port scanning, network services, and vulnerability assessment.**

3. Phishing Email Detection

The phishing email detection project focuses on identifying suspicious emails and classifying them as **Phishing or Safe**. The system analyzes the textual content of emails and uses machine learning techniques to make predictions. The performance of the model can be evaluated using measures such as **accuracy and a confusion matrix**. This project demonstrates the application of machine learning in cybersecurity.

4. Secure Login System

The secure login system focuses on protecting user accounts through secure authentication. It includes user registration, login, password hashing using

bcrypt, input validation, protection against SQL injection, session management, and logout functionality. The project demonstrates important concepts of **web security and secure user authentication**.

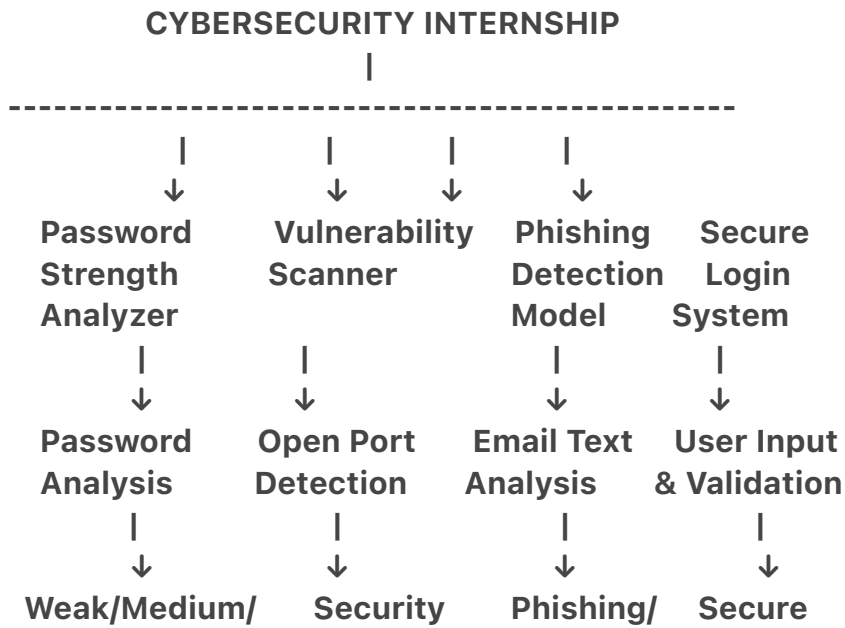
4.6 Technologies and Tools Used

The following technologies and tools were used during the internship:

Technology/Tool	Purpose
Python	Main programming language
Regular Expressions (re)	Password pattern checking
Socket Module	Network communication and port scanning
Pandas	Data handling
Scikit-learn	Machine learning and email classification
Flask	Web application development
SQLite	Database management
bcrypt	Secure password hashing
Visual Studio Code	Code development
GitHub	Project hosting and version control

4.7 System Architecture

The overall system consists of four independent cybersecurity projects.



Strong

Warnings

Safe

Authentication

4.8 Methodology:

The projects were developed using the following methodology:

Step 1: Requirement Analysis

The requirements of each cybersecurity task were studied and the required features were identified.

Step 2: Design

A simple design was prepared for each project based on its objectives and expected outcome.

Step 3: Development

The projects were developed using Python and different libraries.

- Task 1 used regular expressions for password analysis.
- Task 2 used the socket module for port scanning.
- Task 3 used machine learning libraries for email classification.
- Task 4 used Flask, SQLite, and bcrypt for secure authentication.

Step 4: Testing

Each project was tested using different sample inputs to verify its functionality.

Step 5: Result Analysis

The output of each project was analyzed to ensure that the expected functionality was achieved.

4.9 Expected Outcomes

The expected outcomes of the four projects are:

Password Strength Analyzer

The system successfully analyzes passwords and classifies them as **Weak, Medium, or Strong** based on security requirements.

Vulnerability Scanner

The scanner identifies open ports and provides basic security warnings about potentially insecure network services.

Phishing Email Detection

The model successfully classifies emails as **Phishing or Safe** based on textual

content and suspicious patterns. Model performance can be evaluated using **accuracy and a confusion matrix**.

Secure Login System

The secure login system provides password hashing, input validation, SQL injection protection, session management, and logout functionality. These features help reduce unauthorized access and protect user accounts from common attacks.

4.10 Certificates of Completion and Other Communication Mail Proof

The following documents should be attached at the end of the report:

- Internship Offer Letter
- Internship Completion Certificate
- Communication emails related to the internship
- Task submission confirmation, if available
- GitHub repository screenshots or links

Github: https://github.com/krrish1211/2025-29_Om_0995_3rdsem_CSE20

5. BIBLIOGRAPHY / REFERENCES

1. Python Software Foundation. **Python Documentation**.
2. Scikit-learn Documentation. **Machine Learning in Python**.
3. Flask Documentation. **Flask Web Framework**.
4. bcrypt Documentation. **Password Hashing Library**.
5. SQLite Documentation. **SQLite Database System**.
6. GitHub. **Project Repository and Version Control Platform**.